

Building a Power Supply

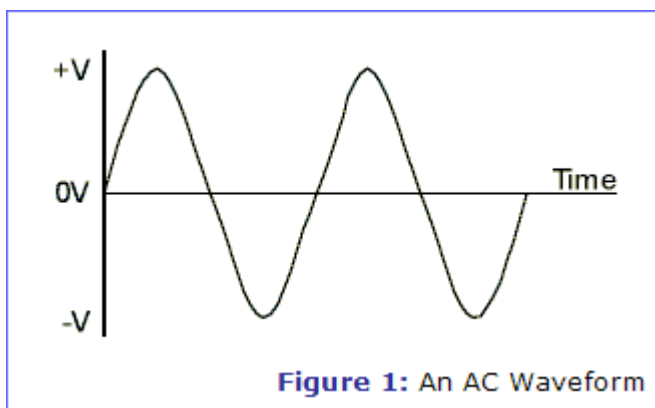
By Tim Surtell

ELECTRONICS
IN MECCANO

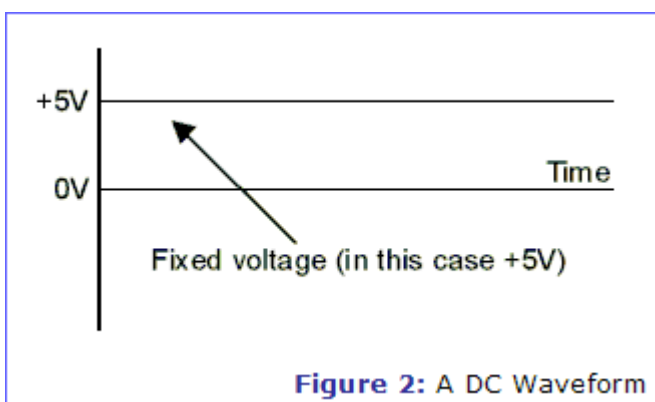
Most of the circuits in *Electronics in Meccano* need a smooth DC power supply in order to function correctly. Some other circuits, particularly those using digital ICs, also need their power supply to be regulated. In this article and the articles that follow in this series you will learn the meaning of terms such as 'smoothing' and 'regulation' and find out how to build a simple power supply for your circuits.

What is AC and DC?

A representation of an Alternating Current (AC) supply is shown in figure 1. The voltage (and current) alternates between positive and negative over time and the resulting waveform shape is a sine wave. In the case of the UK mains supply, the frequency of this sine wave is 50Hz, or 50 cycles per second.



A Direct Current (DC) supply, shown in figure 2, stays at a fixed, regular, voltage all of the time, like the voltage from a battery. A DC supply is needed by most circuits as a constant reference voltage. Also, some components would be damaged by the negative half-cycles of an AC supply.



The Parts of a Power Supply

Figure 3 shows a block diagram of a power supply system which converts a 230V AC mains supply (230V is the UK mains voltage) into a regulated 5V DC supply.

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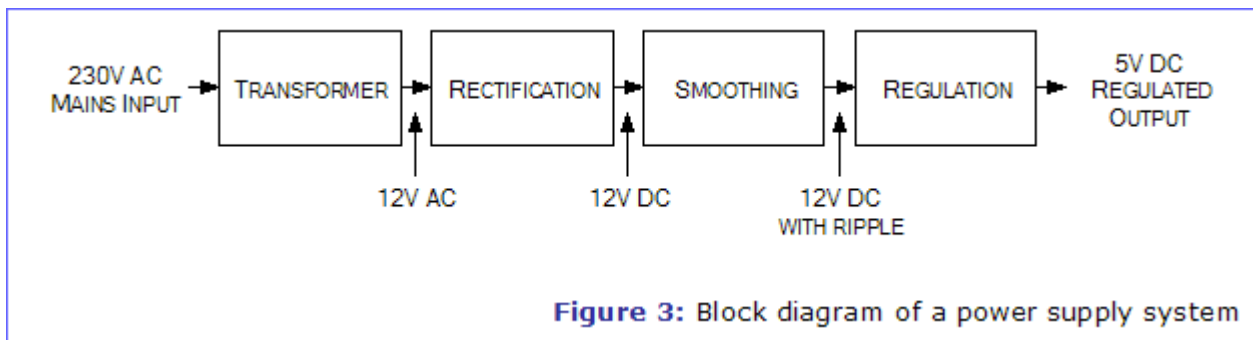
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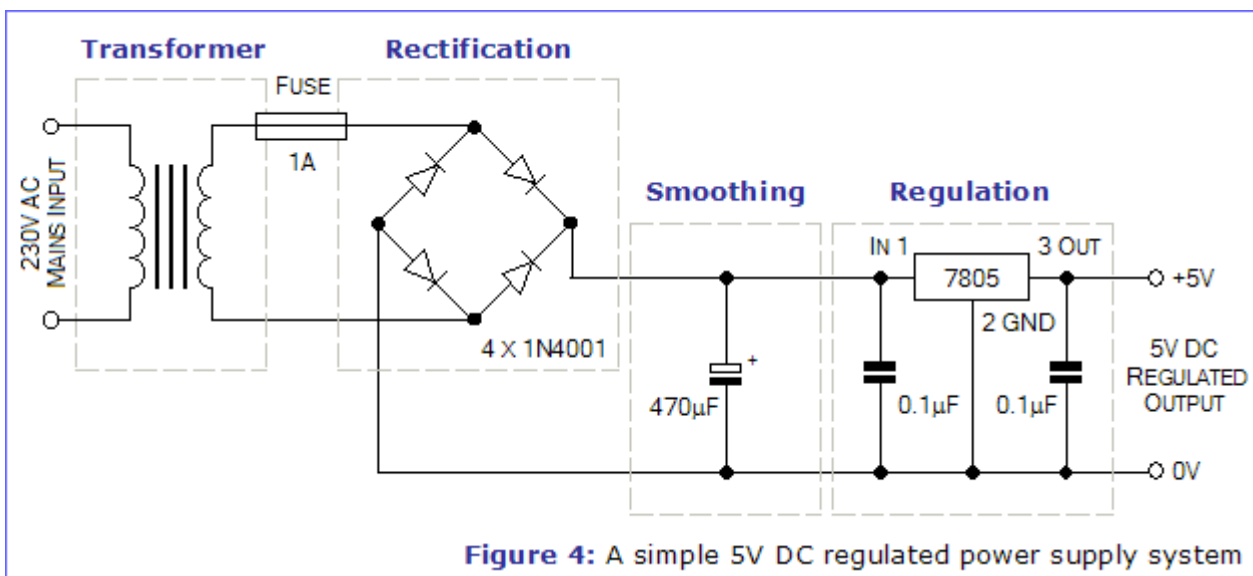
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A simple power supply circuit that includes each of these blocks in given in figure 4. The following articles in this series look at each block of the power supply in detail, but if you just want to build a 5V regulated power supply without understanding how it works, you can follow the instructions later in this article.

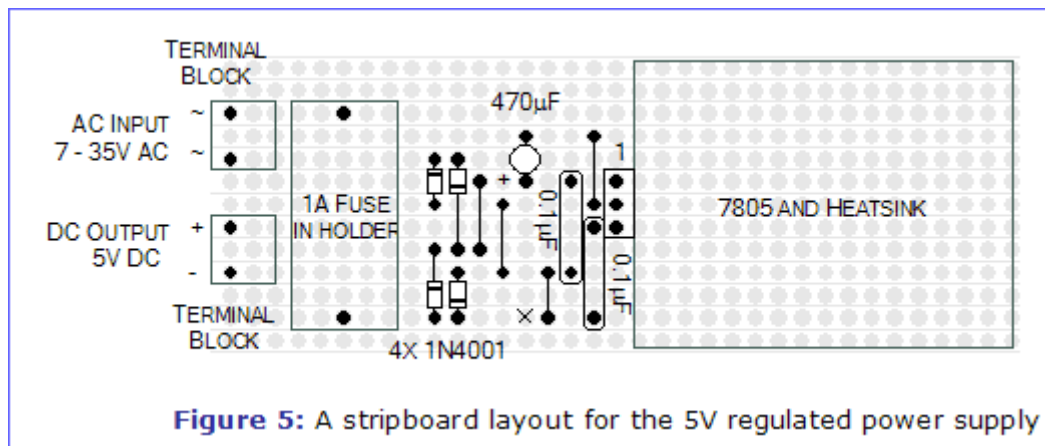
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Building the 5V Regulated Power Supply

Figure 5 gives a stripboard layout for the 5V regulated power supply shown in figure 4. The layout does not include the transformer block, so the input to the board needs to be 7 - 35V AC from a suitable transformer. The layout includes space for two optional 2-way screw terminal blocks to make connecting up the power supply easier.

If the input voltage is 9V AC, you will be able to draw 1A from the power supply. For the maximum input voltage of 35V you will be able to draw 0.1A.



Construction

1. Cut a piece of stripboard to 13 tracks x 35 columns (assuming the use of the 50mm heatsink given).
2. Drill four 1.5mm holes for the thick legs of the terminal blocks, two 2mm holes for the thick legs of the fuse holder, and two 3mm holes for the heatsink and 7805 bolts.
3. Fit the four wire links.
4. Fit the four 1N4001 diodes, taking care that the polarity of each one is correct.
5. Fit the two 0.01µF and the single 470µF capacitors. Ensure that the polarity of the 470µF capacitor is correct. The leads will be marked with '+' or '-'.
6. Fit the two terminal blocks.
7. Fit the fuse holder and insert the fuse. Clip the cover over the fuse holder.
8. Bolt the heatsink to the board.
9. Bend the leads of the 7805 and position it. Bolt it into place before soldering the leads to the board.
10. Cut the copper tracks where an X is shown. Cut any tracks under the heatsink that might short out the 7805 or other stripboard tracks.
11. Connect up a 7V - 35V AC power supply and test the circuit by placing a volt meter across the DC Output terminals. The voltage should read approximately 5V DC.



Shopping List

Please note that the individual nuts, bolts and washers specified to fix the heatsink are no longer available from Maplin. However, they do stock a Nut and Bolt Set containing the specified M2.5 nuts and bolts (no washers).

Part Description	Quantity Required	Maplin Code	Unit Price	Total Price
0.1µF Metallised Polyester Film Capacitor	2	BX76H	£0.20	£0.40
180 Piece Nut and Bolt Set	1	N52BU	£2.26	£2.26
1N4001 Rectifier Diode	4	QL73Q	£0.12	£0.48
20mm 1A Quick Blow Fuse	1	GJ90X	£1.21 (per pack of 10)	£1.21
20mm PCB Mounting Fuse Holder With Cover	1	KU29G	£0.44	£0.44
2-way PCB Mounting Terminal Block	2	RH76H	£0.49	£0.98
470µF 63V Radial Electrolytic	1	VH48C	£0.72	£0.72

Capacitor

50mm Heatsink for TO220 Style Transistor Cases	1	RN79L	£1.51	£1.51
7805 1A Regulator	1	CH35Q	£0.59	£0.59
Stripboard SRBP (29 rows x 39 columns)	1	JP47B	£2.70	£2.70

Grand Total £11.29

► [Full printer-friendly parts list](#)

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